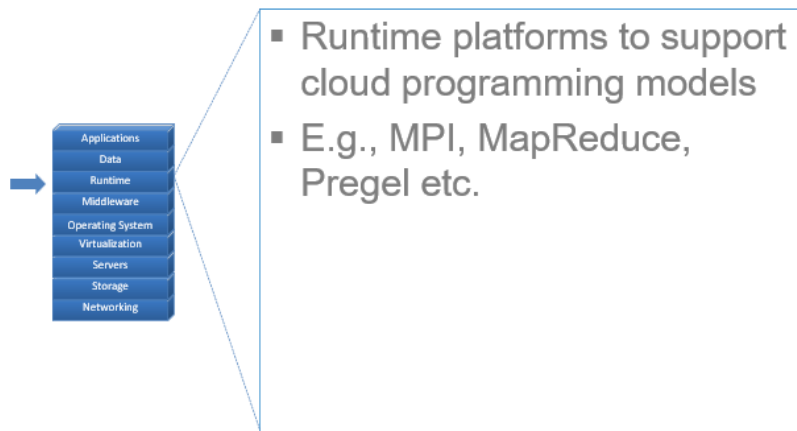


Runtime Environment



1. Five Characteristics of Cloud Computing (As per NIST)

1. **On-Demand Self-Service** – Users can provision resources as needed without human interaction with the service provider.
2. **Broad Network Access** – Services are accessible over the network through standard platforms (e.g., laptops, mobiles).
3. **Resource Pooling** – Provider's resources are pooled to serve multiple consumers using a multi-tenant model.
4. **Rapid Elasticity** – Resources can be quickly scaled up or down based on demand.
5. **Measured Service** – Usage is monitored, controlled, and reported for transparency and billing.

2. Virtualization Security Requirements

- **Isolation** – VMs must be isolated from each other to prevent data leakage.
- **Hypervisor Security** – Secure the hypervisor to prevent unauthorized access or attacks.
- **Access Control** – Strict authentication and authorization policies for VM management.
- **Monitoring** – Continuous monitoring of virtual environments for anomalies or breaches.
- **Patch Management** – Regular updates to hypervisors and virtual systems to fix vulnerabilities.

3. What is the Microsoft Learn Sandbox?

The **Microsoft Learn Sandbox** is a free, temporary environment provided by Microsoft for hands-on learning. It allows users to safely experiment with Azure services without needing a personal Azure subscription. The sandbox expires after a limited time and resets automatically.